

Class 10 • Full Practice Test (Mathematics and Science)

Time: 2½ hours • Maximum marks: 69 • Answer all questions. Show your working.

Section A • Mathematics

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| Q1. | The LCM of 12 and 18 is:
(a) 36 (b) 72 (c) 6 (d) 216 | [1] |
| Q2. | Find the zeroes of the polynomial $x^2 - 5x + 6$ and verify the relationship between the zeroes and the coefficients (sum of zeroes). | [2] |
| Q3. | Solve by the elimination method: $x + y = 7$ and $x - y = 1$. | [3] |
| Q4. | Assertion (A): The equation $x^2 + 4 = 0$ has no real roots.
Reason (R): If the discriminant of a quadratic equation is negative, it has no real roots.
(a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true. | [1] |
| Q5. | Find the 10th term of the AP: 2, 7, 12, ... | [2] |
| Q6. | In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 2$ cm, $DB = 3$ cm and $AE = 4$ cm, find EC. | [3] |
| Q7. | Find the distance between the points (3, 4) and (-1, 1). | [2] |
| Q8. | Prove that $(1 - \cos^2 A) \cdot \operatorname{cosec}^2 A = 1$. | [3] |
| Q9. | A tower casts a shadow 20 m long on level ground when the Sun's angle of elevation is 60° . Find the height of the tower. | [3] |
| Q10. | A point A is 5 cm from the centre O of a circle of radius 3 cm. Find the length of the tangent AP drawn from A to the circle. | [2] |
| Q11. | Find the area of a sector of a circle of radius 7 cm with central angle 60° . (Use $\pi = 22/7$.) | [3] |
| Q12. | A solid toy is in the form of a hemisphere of radius 3.5 cm surmounted by a right circular cone of the same radius and height 12 cm. Find the volume of the toy. (Use $\pi = 22/7$.) | [5] |
| Q13. | Find the mean of the following data:
x: 2, 4, 6, 8 f: 3, 5, 4, 2 | [3] |
| Q14. | A die is thrown once. Find the probability of getting (i) a prime number [2]; (ii) a number greater than 6 [1]. | [3] |

Section B • Science

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| Q15. | Write the balanced chemical equation for the reaction of sodium with water and name the type of reaction. | [2] |
| Q16. | The pH of a neutral solution at room temperature is:
(a) 0 (b) 7 (c) 14 (d) 1 | [1] |

Q17.	Write the balanced chemical equation for the reaction of zinc with dilute hydrochloric acid [2]. How will you test the gas evolved? [1]	[3]
Q18.	Why does carbon form compounds mainly by covalent bonding?	[2]
Q19.	Name the tube that carries food from the mouth to the stomach, and the process by which food moves along it.	[2]
Q20.	What is a reflex action? Trace the path of a reflex arc.	[3]
Q21.	In yeast, asexual reproduction takes place mainly by: (a) budding (b) binary fission (c) fragmentation (d) spore formation	[1]
Q22.	A pure tall pea plant (TT) is crossed with a pure dwarf pea plant (tt). What will be the phenotypic ratio [1½] and the genotypic ratio [1½] in the F ₂ generation?	[3]
Q23.	(a) An object is placed 10 cm in front of a convex lens of focal length 15 cm. Find the position and nature of the image. [3] (b) Draw the ray diagram for this case. [2]	[5]
Q24.	Why does the clear sky appear blue?	[2]
Q25.	A copper wire is 2 m long and has an area of cross-section $1 \times 10^{-6} \text{ m}^2$. The resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$. Find its resistance.	[3]
Q26.	State Fleming's left-hand rule.	[2]
Q27.	Case-based: In a grassland, the food chain is: grass → grasshopper → frog → snake. (i) At which trophic level is the frog? [1] (ii) If the grass traps 10,000 J of energy, how much energy reaches the snake (10% law)? [1] (iii) What will happen to the grasshopper population if all the frogs are removed? Give a reason. [2]	[4]